

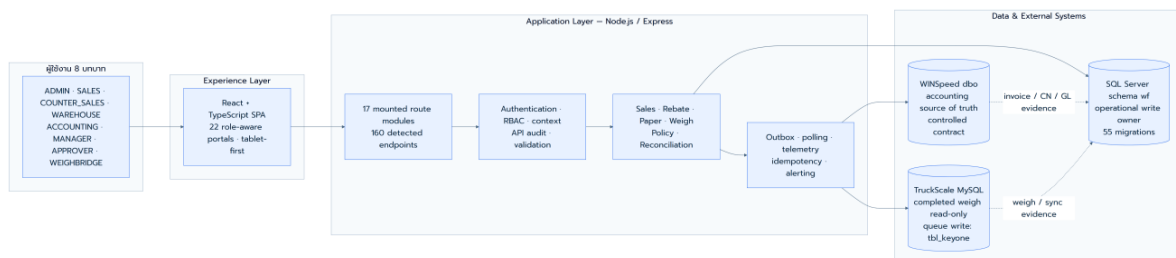
Analysis and Design Document

Stakeholders, Architecture, Decisions, Data and Workflows

SOURCE-ALIGNED CANDIDATE · REVIEW

Control	Value
Document ID	WF-SAD-001
Version / Runtime	v1.0 candidate / runtime 1.0.1
Date	23 July 2026 / 23 กรกฎาคม 2569
Source commit	79a10a28e6a2fba9b65dc85101ff8ab6d784b91c
Source inventory SHA-256	12B9F964C7C90341859EDD6CDDE9B92BA35D797347F2AA64A1134E1E885FC343
Automated evidence	2026-07-23T09:56:59-217Z · 10/10 passed · source stable
Status	Review — approval and baseline acceptance required

การใช้เอกสาร สร้างจาก Markdown และ source evidence ล่าสุดตามนโยบาย latest-document-wins; ห้ามตีความเป็น Approved, Released หรือ ISO certification จนกว่าจะมีผู้มีอำนาจลงนามและรับ baseline.



Executive visual — Analysis and Design Document

Document control

รายการ	รายละเอียด
Document ID	WF-SAD-001
Title	Analysis and Design Document
Owner	Solution Architect / Business Analyst / Data Architect
Status	Review — Candidate
Classification	Confidential — Client / Authorized Partner Use Only
Design preset	standard_business_brief · A4 · Prompt/Kanit override
Header pattern	memo_masthead
Generation rule	drift check → source merge → render → QA → human review

Revision and approval record

Version	Date	Change	Prepared by	Reviewed / approved
v1.0 candidate	2026-07-23	Rebuilt from source-aligned Markdown, runtime inventory, diagrams and current E2E evidence	Documentation pipeline	Pending named human approval

Executive source snapshot

Evidence area	Current verified fact	Boundary
Runtime	10.1 across root/backend/frontend packages	Development source snapshot
Source	220 files · SHA 12B9F964...FC343 · commit 79a10a28	Dirty worktree remains visible
Application	22 portals · 8 roles	Role-specific UAT still required
API	17 route mounts · 160 detected endpoints	Detection is not contract approval
Data	55 sequenced migrations through 055	Production migration/restore gate remains
External writes	33 detected dbo writes · 2 TruckScale writes to tbl_keyone	Explicit contract/security review required
Automated E2E	10/10 passed · stable source · SQL Server/MySQL up	Not production hardware/manual UAT

Evidence interpretation ตัวเลขจาก static inventory และ E2E ใช้เพื่อยืนยันความสอดคล้องกับ snapshot ปัจจุบันเท่านั้น การยอมรับ UAT, external write boundary, production readiness และ ISO conformance ต้องมีหลักฐานและผู้อนุมัติแยกต่างหาก.

Standards alignment

Reference	Use in this information set	Current document
ISO/IEC/IEEE 42010:2022	Stakeholders, concerns, viewpoints, views and decisions	Primary

WS-Sale-App Analysis and Design Document		WF-SAD-001 · REVIEW
Reference	Use in this information set	Current document
ISO/IEC/IEEE 12207:2026	Life-cycle processes, reviews and evidence flow	Development information set
ISO/IEC/IEEE 15289:2019	Information-item content and document control	Front matter / source register
ISO/IEC 25010:2023	Product quality characteristics and verification concerns	NFR / architecture / technical gates
ISO/IEC/IEEE 29119-3:2021	Test documentation, cases, results and traceability	Verification and UAT linkage

Audience and reading path

Audience	Primary concern	Recommended sections
Business/process owner	stakeholders and cross-role process	Business analysis, workflow, decisions
Architect / developer	boundaries, components and constraints	SAD, C4, ADR, UML
DBA / integration owner	ownership, data and external contracts	Data design, ERD, integration decisions
Operations / audit	risk, evidence and maintainability	Architecture concerns, decision records

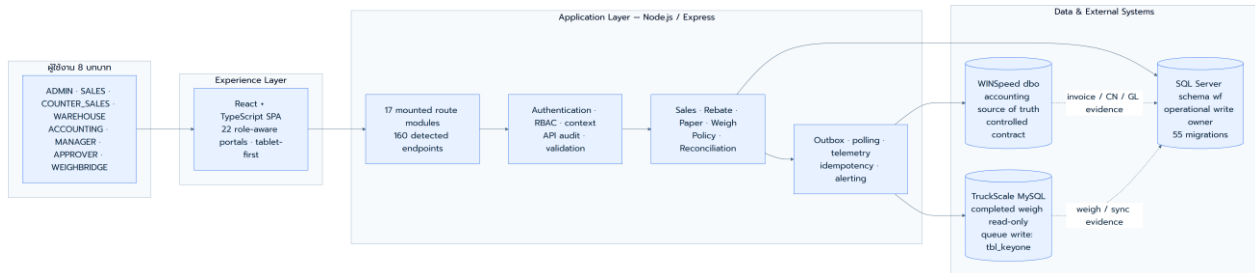
Table of contents

Update field in Word to refresh the table of contents

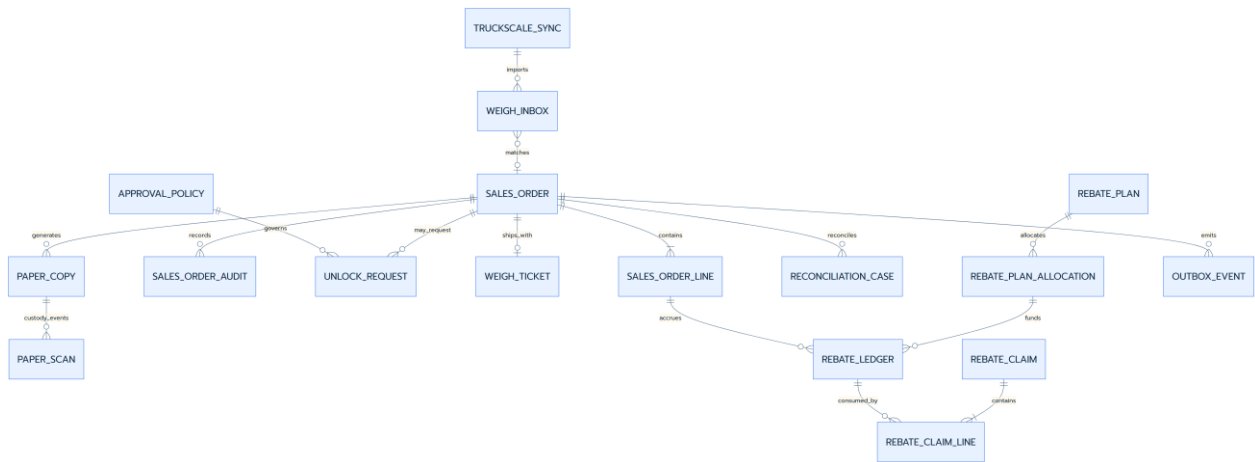
Word field เปิดเอกสารใน Microsoft Word แล้วกด Ctrl+A และ F9 เพื่อ refresh สารบัญและจำนวนหน้า.

Source-aligned diagrams

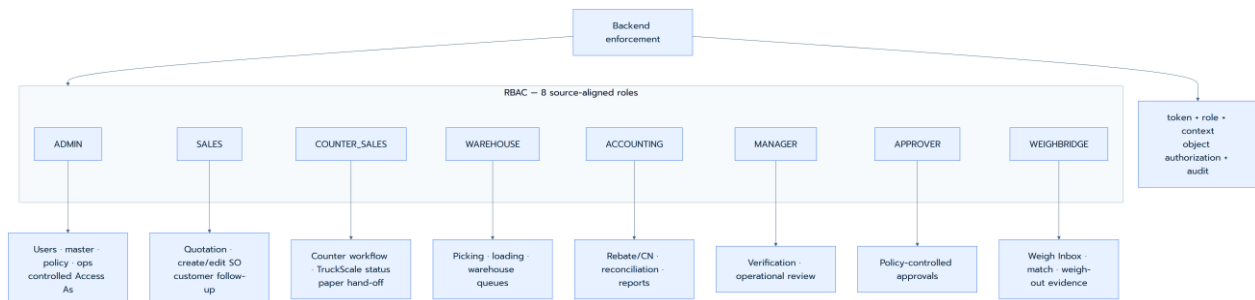
Diagram policy ภาพในส่วนนี้สร้างจาก Mermaid source ที่ version-control ได้และ render ใหม่ด้วย Prompt/Kanit policy; แหล่ง Markdown ยังคงเป็นข้อมูลอ้างอิงและภาพต้องผ่าน visual QA ก่อน release.



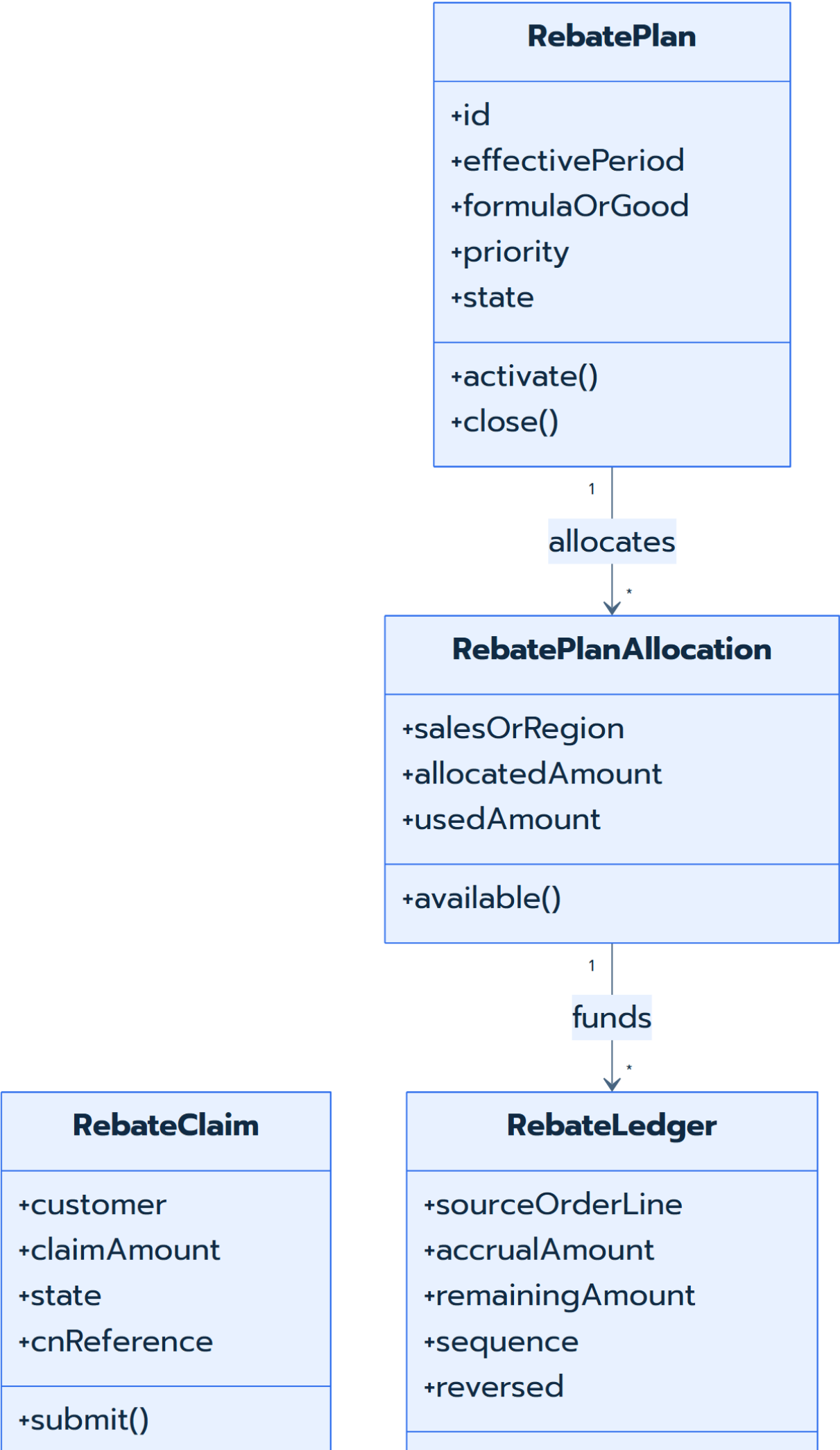
ภาพที่ 1 – Runtime architecture and system boundaries



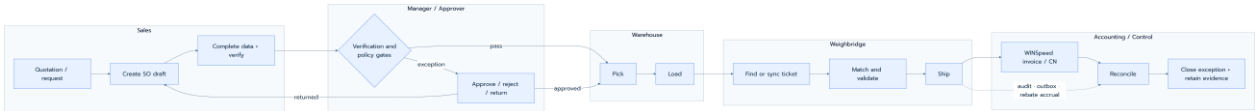
ภาพที่ 2 – Operational ERD and evidence relationships



ภาพที่ 3 – Source-aligned RBAC responsibility map



תהליך 4 – Rebate domain UML



תהליך 5 – Cross-role order-to-cash workflow

Business Analysis Document – Sales Execution & Rebate Platform

Authoritative source docs/enterprise/01-BUSINESS-ANALYSIS/BAD-v8.md · SHA-256 1DEB832C9B5D1B425CF369720BBAF04955E934B1F5AB2382E7473F7BEDA2DD71 · merged as Review content

Executive statement

World Fert ต้องการลดงานคีย์เข้า ลดเอกสารสูญหาย ควบคุมการจ่ายสินค้าให้ตรงคำสั่งขาย และทำให้ rebate ตรวจสอบย้อนหลังได้ระดับรายการ โดยไม่กระทบการลงบัญชีเดิมของ Prosoft WINSpeed v9.0

WS-Sale-App จึงเป็น Sales Execution & Control Layer ไม่ใช้การทดแทน ERP ทั้งหมดในระยะแรก

Business context

- บริษัทผลิตและจำหน่ายปุ๋ยหลายสูตร ขายเป็นตันและรับสินค้าในโรงงาน
- คำสั่งซื้อผ่านฝ่ายขายและ Counter Sales ก่อนเข้าสู่คลัง/โรงงาน
- รถอวรับสินค้าหลายจุด เช่น สายพาน คลัง และ bulk
- Rebate ตั้งตามสูตร พื้นที่ ระยะเวลา และราคา NET แต่การเคลมเดิมตรวจสอบเอกสารมาก
- WINSpeed ต้องคงความเป็นเจ้าของเอกสารบัญชีและ GL

AS-IS process

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Pain points ที่ยืนยันแล้ว

ID	ปัญหา	ผลกระทบ	เป้าหมาย v7
BP-01	คีย์ order เข้าจากแชตไป ERP	ช้าและเกิด error	POS/tablet-first, validation, draft
BP-02	เอกสาร 4 สืบติดตามยาก/สูญหาย	audit gap	QR, paper trail, alert
BP-03	แก้ SO หลังเริ่มจ่ายสินค้าไม่มี workflow	audit หาย	Unlock + reversible ledger
BP-04	Rebate ตรวจสอบทีละ invoice	งาน Accounting สูง	ledger line-level + FIFO
BP-05	ตัด rebate แบบก้อนรวม	trace ต้นทางไม่ได้	pool + ledger + allocation
BP-06	ราคา NET/promotion เปลี่ยนบ่อย	execution ไม่ทัน	Price Book + Plan version
BP-07	คลังตีความลำดับแม่/ลูก	โหลดผิด/ช้า	load sequence ต่อ line
BP-08	ของแถมอยู่ในหมายเหตุ	ไม่เห็นงบ/สต็อก	giveaway workflow
BP-09	ตัวคุมรายชุดดูยาก	ตรวจสอบยอดยาก	drill-down balance
BP-10	TruckScale แยกแยะ data quality ไม่สม่ำเสมอ	integration risk	completed-weigh read-only + controlled tbl_keyone pre-weigh contract/match policy

To-Be operating model

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Business outcomes and KPI

Outcome	KPI
ลดเวลาคลี่/ตรวจ SO	median create-to-confirm time, rework rate
ลดการ rebate	claim-review time, % traceable automatically
ลดเอกสารสูญหาย	open/lost paper-copy count and aging
ควบคุมการจ่าย	% shipped SO with verification, weight and audit
รักษามาตรฐาน	reconciliation pass rate with WINSpeed
ควบคุม promotion	plan budget variance / giveaway exception
ตัดสินใจเร็ว	dashboard freshness / queue aging

Scope baseline

In scope

1. Sales order, quotation-to-SO and Verification workflow
2. Mother/baby load sequencing
3. Rebate plan, allocation, accrual, FIFO, claim and CN evidence
4. Giveaway budget/withdrawal
5. Control-ticket balance and report
6. Weigh-out capture, TruckScale lookup and manual fallback
7. Paper Trail, QR and audit
8. Dashboard, reports, RBAC, security and operations

Out of scope unless separately approved

- full factory production/MRP
- Gate Pass replacement
- TruckScale hardware/serial controller replacement
- full LINE LIFF application
- lot/expiry traceability
- changes to WINSpeed GL/accounting logic

Critical decision gates

Gate	Decision required	Owner
DG-01	WINSpeed write boundary	Executive + IT + Accounting
DG-02	Canonical TruckScale source	IT + Factory
DG-03	Credit Hold: build/defer	Sales + Accounting
DG-04	Stock source and reservation method	Warehouse + Finance
DG-05	Approval authority thresholds	Management
DG-06	Privacy/document retention	Management + QMR

Stakeholder Map, RACI and Approval Authority

Authoritative source docs/enterprise/01-BUSINESS-ANALYSIS/RACI-AND-STAKEHOLDERS.md · SHA-256 52D22376C6B1D1CC28DCF1D6DAB6F872970894A137313C7FEF52F206A9395798 · merged as Review content

Roles

Role	Responsibility
SALES	สร้าง SO, rebate/claim, request unlock
COUNTER_SALES	verify SO, print/control documents
WAREHOUSE	picking/load sequence
WEIGHBRIDGE	weight/source validation/ship
APPROVER	unlock/exception/plan approval
ACCOUNTING	claim/CN/reconciliation
ADMIN	users/master/configuration
MANAGER	plan/budget/report policy
IT/DBA	infrastructure/security/backup
EXECUTIVE	scope, high-risk change, go-live

RACI

Activity	Sales	Counter	Warehouse	Weigh	Accounting	Admin/IT	Manager	Exec
Create SO	R	A/C	I	I	I	I	I	I
Verify SO	C	A/R	I	I	I	I	I	I
Confirm/exception	R	C	I	I	C	I	A	I
Pick/load	I	C	A/R	C	I	I	I	I
Weigh/ship	I	I	C	A/R	I	I	I	I
Rebate Plan	C	I	I	I	C	I	A/R	I
Claim/CN	R	I	I	I	A/R	I	C	I
Security/backup	I	I	I	I	I	A/R	C	I
Go-live	I	C	C	C	C	C	R	A

R=Responsible; A=Accountable; C=Consulted; I=Informed

Approval baseline

Scenario	Required approval
Unlock after Picking	authorized Approver

WS-Sale-App Analysis and Design Document		WF-SAD-001 · REVIEW
Scenario	Required approval	
Sell above set price	named authority by policy	
Sell below allowed threshold	authorized manager matrix	
Price activation	Manager + Accounting review	
Rebate plan activation	Manager/Approver	
Overbudget giveaway	manager/delegate + reason	
Manual weight override	Weighbridge + supervisory review	
Direct dbo exception	IT/DBA + Accounting + delegated Executive	

Solution Architecture Document

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/SAD-v8.md · SHA-256 751BD422F3E4C39DC4E43A6B8572CC48D87351775A5A4617D36FE5CA2CC49777 · merged as Review content

Architecture intent

ออกแบบเพื่อให้ operational workflow เปลี่ยนเร็วโดยไม่กระทบ finance/GL ของ WINSpeed และสร้างหลักฐานที่ตรวจสอบได้ระดับ event/record

Logical architecture

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Architecture principles

9. Accounting boundary — WINSpeed owns invoice/CN/GL posting.
10. Controlled write boundary — direct dbo writing is explicit exception, never implicit.
11. Append-only evidence — audit, ledger and source references preserved; corrections compensate.
12. Read-model isolation — heavy report/control-ticket queries use approved views/indexes/pagination.
13. Integration resilience — idempotency, retry/reconciliation and manual fallback explicit.
14. Least privilege — no sa/root in runtime.
15. Observe before mutate — completed-weight TruckScale records are read-only; controlled pre-weight queue writes are limited to tbl_keyone; ambiguity is surfaced.
16. Configuration over hardcode — policy/threshold/period/environment controlled.
17. Operational transparency — health, version, release, failures visible.

Bounded contexts

Context	Owned data	Key responsibility
Sales Execution	SO, lines, states, verify, load sequence	lifecycle control
Rebate	plan, allocation, pool, ledger, claim	traceable promotion
Warehouse/Weighing	pick status, WeighTicket	loading/ship evidence
Paper Trail	copies, QR, scans	document custody
Integration	outbox/reconcile	safe external operations
Identity/Admin	user/role/policy	access/config
Reporting	read models/export	operational visibility

Data ownership

Data class	Owner	App authority
WINSpeed master	WINSpeed	controlled read
WINSpeed invoice/CN/GL	WINSpeed	read/reconcile
operational workflow/audit	wf	write
Rebate/claim	wf + CN evidence	write/read

WS-Sale-App Analysis and Design Document		WF-SAD-001 · REVIEW
Data class	Owner	App authority
Weigh source	TruckScale	completed records read-only; tbl_keyone pre-weigh queue write
Paper custody	wf	write

Integration sequence

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Production topology target

- Frontend through managed static hosting/CDN or on-prem Nginx
- Backend as immutable container
- SQL Server via private network/VPN/allowlist
- TruckScale via private connectivity or managed replica
- centralized secret management
- monitoring/alerting/backup validation before Full Production

Current implementation view

View	Source-aligned fact
Runtime	frontend, backend and root packages declare version 1.0.1
Application surface	22 portal keys, 8 roles, 17 mounted API route modules and 160 detected endpoints
Persistence	SQL Server wf operational extension, controlled WINSpeed dbo, TruckScale MySQL
Schema evolution	55 sequenced migration files through sequence 055 with no duplicate sequence
Controlled external writes	33 detected dbo write statements require contract review; TruckScale writes are limited to 2 statements targeting tbl_keyone
Automated evidence	E2E run 2026-07-23T09:56:59-217Z: 10/10 passed, source stable, SQL Server/MySQL health up

Architecture concerns and verification gates

- Production connectivity, scale hardware, printing, QR scanning and manual recovery remain manual/integration test gates.
- Direct WINSpeed writes are exceptions governed by ADR-003 and the approved integration contract; detection count is an inventory signal, not an approval.
- Deployment secrets, backup restore, alert delivery and network failover require environment-specific evidence before go-live.
- Architecture descriptions follow ISO/IEC/IEEE 42010:2022 concepts: stakeholder concerns, viewpoints, views, decisions and correspondence to implementation evidence.
- Life-cycle deliverables and reviews are organized against ISO/IEC/IEEE 12207:2026 and information-item control follows ISO/IEC/IEEE 15289:2019; these references do not by themselves certify the solution.

C4 Architecture Views

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/C4-ARCHITECTURE.md · SHA-256 101BACF722B8A600204A0B0106779C518BF0F43692D2A1CC6013EC1EB3DA4FD7 · merged as Review content

Level 1 — System Context

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Level 2 — Containers

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Level 3 — Backend components

Component	Responsibility	Boundary
Auth middleware	token/RBAC/context	protected endpoints
Sales service	lifecycle, verification, audit	wf + controlled contract
Rebate service	plan/pool/ledger/FIFO/claim	wf
Weigh service	lookup/normalize/ticket persistence	TruckScale read + wf
Paper service	copies/QR/scans/custody	wf/storage
Integration service	outbox/import/reconcile/idempotency	WINSpeed + wf
Reporting service	read models/export	views/read models
Admin service	users/roles/policy/master	wf/contracts
Health/telemetry	dependency status/error metrics	runtime

Level 4 — Code constraints

- routes must not hold financial/business rules that belong in services
- every mutable workflow emits audit event through shared service
- parameterized SQL only
- cross-database calls require timeout and explicit contract
- write operations support idempotency where retry can duplicate outcome

ADR-001 — Layered Non-Invasive Integration

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-001-LAYERED-NONINVASIVE.md · SHA-256
DDD206566D005742700117157341B40C9BBD1435E609CF3A693B50C0804B0C82 · merged as Review content

Context

WINSpeed is business-critical and owns historical master, finance and GL. Replacement creates unacceptable operating/accounting risk.

Decision

WS-Sale-App operates as operational extension:

- workflow/audit/rebate/paper/weigh data in wf
- WINSpeed dbo through approved read contracts
- external writes under ADR-003
- fallback remains available through WINSpeed/manual procedure

Consequences

Positive: lower replacement risk, direct data visibility where permitted, faster operational improvements.

Negative: contracts/reconciliation mandatory; legacy data quality remains visible; raw table access is not a design substitute.

Guardrails

- no schema modification to WINSpeed
- no direct GL/invoice/CN posting
- coexistence/reconciliation test at release gate

ADR-002 — Three-Database Topology and Environment Control

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-002-DATABASE-TOPOLOGY.md · SHA-256 7001AB6065F504C91B5CD841061AFE7BD6FE076166576C72EDE720244B9E6A62 · merged as Review content

Context

The solution uses SQL Server WINSpeed dbo, SQL Server wf schema, and TruckScale MySQL. Evidence shows wf/dbo share SQL Server while TruckScale is logically separate.

Decision

- wf is write-owned operational schema
- dbo is legacy/external schema controlled by ADR-003
- TruckScale uses a dedicated service/pool: completed-weigh/reference tables are read-only and pre-weigh queue writes are restricted to tbl_keyone
- environment selection propagates through explicit request context
- production environment register names host, DB, runtime identity, owner, backup policy and data class for each source

Consequences

No cross-engine join is assumed at runtime. Cross-source correlation is resolved through application/service logic or an approved read model.

ADR-003 — WINSpeed Write Boundary and Financial Safety

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-003-WINSPEED-WRITE-BOUNDARY.md · SHA-256 42316FD65943B863B07F10908E11050644961DB9368466FCE9D9B29EE194557A
· merged as Review content

Problem

Existing implementation evidence includes direct writes to selected WINSpeed objects for SO/picking/ship/price. The target production safety model requires WINSpeed to own invoice/CN/GL and its posting/import process.

Decision

Immutable financial boundary

WS-Sale-App MUST NOT directly insert/update:

- SOInvHD, SOInvDT
- GLHD, GLDT
- other financial-posting objects not explicitly approved by WINSpeed DBA and Accounting

Approved write paths

Use case	Preferred path	Temporary fallback	Required controls
Create/Update SO 104	WINSpeed import or approved stored procedure	legacy direct write only with signed exception	idempotency, transaction, audit, dual-run
Picking status	approved interface/SP	signed direct-update exception	audit/state validation
Ship/export state	controlled interface/import	signed direct-update exception	weigh evidence/reconcile
Price Book	approved maintenance contract	direct EMSetPrice* under policy	dual approval/effective date/audit
Invoice/CN/GL	WINSpeed UI/import/post	none	Accounting owns posting

Legacy exception exit plan

18. Inventory direct writes.
19. Risk-classify by object/consumer.
20. Create import/SP contract.
21. Dual-run/reconcile 2–4 weeks.
22. Disable direct write via feature flag.
23. Retain evidence.

Go-Live criterion

No production release uses undocumented direct dbo write. A temporary exception lists object, columns, role, approver, expiry, rollback and reconciliation query.

ADR-004 — TruckScale Controlled Pre-Weigh Integration and Matching Policy

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-004-TRUCKSCALE-INTEGRATION.md · SHA-256 DE63608F9127325F6F68F319B607FF71857BF63443FEE36BC932F630BBDBC0DF · merged as Review content

Context

TruckScale records pre-weigh, completed weighing and product detail, but has Thai string/serial dates, mixed charsets, no enforced FK, inconsistent customer codes and imperfect truck-plate values.

Decision

24. TruckScale completed-weigh and reference records are read-only; WS-Sale-App may INSERT/DELETE only tbl_keyone for the pre-weigh queue lifecycle.
25. Dedicated connection/service has timeout, health and charset handling.
26. Candidate match uses normalized plate as primary; movebill, document/invoice and time window corroborate.
27. Candidate evidence/confidence is visible; ambiguity requires user selection.
28. Selection is copied as immutable reference metadata to wf.WeighTicket.
29. Production environment register declares canonical source: local, cloud replica, or managed integration DB.

Matching policy

Step	Rule
Normalize	trim spaces/punctuation; preserve raw
Filter	same normalized plate + date window
Rank	completed record > matching movebill > matching source doc > closest time
Validate	net>0, gross>tare, capacity sanity, no duplicate ref
Resolve	ambiguous candidate selected by user + audit
Fallback	manual entry authorized + reason + later reconcile

Non-goal

WS-Sale-App does not control COM ports/hardware or alter completed TruckScale source records; the controlled tbl_keyone queue lifecycle is the only write exception.

ADR-005 — Rebate Ledger, FIFO and Compensation Model

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-005-REBATE-FIFO.md · SHA-256 1C0C516C879AC8E39AFFAFDFF49B775A28794E33C45DF270E7E474454328F5C4 · merged as Review content

Decision

Rebate is represented as immutable ledger events, not one mutable aggregate balance.

- confirmed eligible SO line creates accrual using effective plan/NET snapshot
- pool is controlled aggregation, not sole source of truth
- claim consumes eligible accrual FIFO
- unlock/reversal creates compensating event/flag, never deletion
- claim approval links actual WINSpeed CN evidence
- price-difference return and sales rebate are distinct types

Benefits

Every claimed amount can trace to SO line, plan, price and date. This supports audit and prevents unexplained remaining-pool behavior.

Data Design, Ownership and Canonical Model

Authoritative source docs/enterprise/04-DATA-INTEGRATION/DATA-DESIGN.md · SHA-256 02C1218BA62552576952B01A7ECDC7BA24741919B6174907C6620059213AF243 · merged as Review content

Data ownership

Domain	Primary owner	Write mode	Retention/audit
Customer/good/account master	WINSpeed	controlled external	WINSpeed policy
SO operational workflow	wf	app owned	target 7 years
Rebate/claim	wf + CN reference	app owned / external evidence	target 7 years
Price Book	WINSpeed/source contract	controlled	versioned
Weigh source	TruckScale	tblscale/reference read-only; tbl_keyone controlled queue write	source policy + wf ref
Paper copies/scans	wf	app owned	target 7 years
User/role mapping	wf	admin controlled	access policy
Outbox/reconcile	wf	app owned	until resolved + policy

Logical entities

Sales execution

Entity	Key fields	Notes
SalesOrder	Sold, SoNo, CustomerId, Status, TruckPlate, VerifiedBy, VerifiedAt	header
SalesOrderLine	Sold, LineNo, GoodId, QtyTon, UnitPrice, LoadSequence	commercial/warehouse line
SalesOrderAudit	actor, action, before/after JSON, IP, time, correlation	append-only
UnlockRequest	Sold, reason, status, requester, approver	correction
WeighTicket	Sold, source ref, tare/gross/net, scale, time, method	ship evidence

Rebate and giveaway

Entity	Key fields	Notes
PriceBook	formula/good, month, NET, version, state	effective price
RebatePlan	dates, formula, region, return type, priority	policy
RebatePool	sales, plan, allocated/used/frozen	aggregation
RebateLedger	source line, pool, type, amount, FIFO, reversed flag	subledger
RebateClaim	customer, sales, totals, state, CN number	request

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Entity	Key fields	Notes
RebateClaimLine	claim/ledger, amount, return type	trace
GiveawayBudget	region/sales/item/brand/period	budget
GiveawayWithdrawal	SO/budget/item/qty/reason	issue

Controls and platform

Entity	Key fields	Notes
PaperCopy	SO, color, QR nonce, status, holder	custody
PaperScan	copy, status, actor, timestamp	append-only
AppUser	identity, role, EmpID, active	named account
ApprovalPolicy	scenario, threshold, authority, effective dates	v7 target
IntegrationOutbox	type, idempotency key, payload hash, state	v7 target
ReconciliationCase	entity/ref, expected/actual, state, owner	v7 target

Data constraints

- money uses decimal and explicit rounding; no floating point
- quantity stores canonical ton/kg plus UOM conversion
- datetime uses explicit timezone/UTC policy
- raw and normalized data both retained for matching/search
- audit JSON includes schema/version
- PII is not written to general logs/export by default

Conceptual ER

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

11 – Workflow and Workprocess Diagrams

Authoritative source docs/enterprise/03-SOLUTION-ARCHITECTURE/WORKFLOW-DIAGRAMS.md · SHA-256 F9BC621DB5847A2EE61E616F963946B183B3C7C74E796EECA4AD14C8DAE16054 · merged as Review content

Source note World Fert · WS-Sale-App — Enterprise Documentation v1.0
Document ID: WF-ARCH-010 · Version: v1.0 · Date: 21 מרחשון 2569 (21 July 2026) · Status: Released
Classification: Confidential — Client / Authorized Partner Use Only
Source of truth: operational build v1.0 · verified against dbwins_worldfert9

This document contains structured data in Mermaid.js (<https://mermaid.js.org/>) format.

How to use with Draw.io

30. Open draw.io (<https://app.diagrams.net/>).
31. From the top menu, select Arrange > Insert > Advanced > Mermaid...
32. Copy the Mermaid code blocks below (excluding the mermaid backticks) and paste them into the text box.
33. Click Insert to generate the flowchart instantly.

1. End-to-End Sales Order Flow

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

2. Quotation Conversion Process

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

3. Giveaway Approval Flow

Diagram source Mermaid source is retained in Markdown; the reviewed high-resolution rendering is embedded in the figure section.

Source and evidence register

Type	Path	SHA-256	Disposition
Markdown	docs/enterprise/01-BUSINESS-ANALYSIS/BAD-v8.md	1DEB832C9B5D1B425CF369720BBAF04955E934B1F5AB2382E7473F7BEDA2DD71	Merged into Review candidate
Markdown	docs/enterprise/01-BUSINESS-ANALYSIS/RACI-AND-STAKEHOLDERS.md	52D22376C6B1D1CC28DCF1D6DAB6F872970894A137313C7FEF52F206A9395798	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/SAD-v8.md	751BD422F3E4C39DC4E43A6B8572CC48D87351775A5A4617D36FE5CA2CC49777	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/C4-ARCHITECTURE.md	101BACF722B8A600204A0B0106779C518BF0F43692D2A1CC6013EC1EB3DA4FD7	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-001-LAYERED-NONINVASIVE.md	DDD206566D005742700117157341B40C9BBBD1435E609CF3A693B50C0804B0C82	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-002-DATABASE-TOPOLOGY.md	7001AB6065F504C91B5CD841061AFE7BD6FE076166576C72EDE720244B9E6A62	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-003-WINSPEED-WRITE-BOUNDARY.md	42316FD65943B863B07F10908E11050644961DB9368466FCE9D9B29EE194557A	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-004-TRUCKSCALE-INTEGRATION.md	DE63608F9127325F6F68F319B607FF71857BF63443FEE36BC932F630BDBDCODF	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/ADR-005-REBATE-FIFO.md	1COC516C879AC8E39AFFAFDFF49B775A28794E33C45DF270E7E474454328F5C4	Merged into Review candidate
Markdown	docs/enterprise/04-DATA-INTEGRATION/DATA-DESIGN.md	02C1218BA62552576952B01A7ECDC7BA24741919B6174907C6620059213AF243	Merged into Review candidate
Markdown	docs/enterprise/03-SOLUTION-ARCHITECTURE/WORKFLOW-DIAGRAMS.md	F9BC621DB5847A2EE61E616F963946B183B3C7C74E796EECA4AD14C8DAE16054	Merged into Review candidate
Figure	docs/enterprise/04-DIAGRAMS-PNG/candidate/domain/13-runtime-architecture.png	B08C8F0563C3A84D538E52E28C74370BCDAF05744B2F358CFB60D90319D9E8D3	Embedded with alt text
Figure	docs/enterprise/04-DIAGRAMS-PNG/candidate/domain/15-operational-erd.png	21C695D3CF235B7CE22C49E1ACC3E97A9142D959D95A24BE0FD71A4BEE127D01	Embedded with alt text
Figure	docs/enterprise/04-DIAGRAMS-PNG/candidate/domain/16-rbac-source-aligned.png	A2FAF61E5FCB4E1EC0CB7DAC363119B89073677B1BE188DCD41CB9A1COC78B4D	Embedded with alt text
Figure	docs/enterprise/04-DIAGRAMS-PNG/candidate/domain/18-rebate-domain-uml.png	481932EABE7CE15D8A4E76F25DD2D675002514F7ED99542B6DBED19AA9C4CCAE	Embedded with alt text
Figure	docs/enterprise/04-DIAGRAMS-PNG/candidate/domain/17-order-to-cash-workflow.png	CC2288765EAE21DC86B1B927202EB2DA941677A6623579E16EC6C791555AC207	Embedded with alt text

Approval boundary

Pending human decision Candidate นี้ยังไม่เป็น baseline. ต้อง review ความถูกต้องของ business rule, API/data contract, external write boundary, security, operations และ UAT evidence ก่อนบันทึกผู้อนุมัติ/วันที่/ข้อยกเว้น.

Approval role	Name	Decision / date	Conditions or evidence reference
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WS-Sale-App Analysis and Design Document			WF-SAD-001 · REVIEW
Approval role	Name	Decision / date	Conditions or evidence reference
Business owner / Product owner		Pending	
Solution architect / Technical lead		Pending	
Security / DBA / Integration owner		Pending	
QA / UAT owner		Pending	
Operations / Release authority		Pending	